Vardaan Bajaj

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Experience

DRDO Hyderabad
Software Developer — *Jan 2026 – Present*

- Designed and built software for RS-422, UART and MIL-STD-1553 models.

AgryBin
AI Intern — *May 2025 – Aug 2025*

- Designed and optimized AI pipelines for crop health assessment using satellite imagery, achieving **17% improvement in accuracy**.
- Implemented segmentation and classification models in **PyTorch**, processing **50K+ geospatial image tiles** with GPU acceleration.
- Deployed scalable inference APIs via **Flask + FastAPI**, maintaining **<120ms latency** and enabling automated analytics dashboards.

Mahyco
Data Science Intern — *Aug 2024 – Dec 2024*

- Developed a **YOLO-based crop yield estimation pipeline** using drone imagery, processing **20K+ field images** with preprocessing and real-time detection.
- Built interactive **analytics dashboards** to visualize spatial yield and detect anomalies, improving decision-making by 23%.
- Integrated backend services with **Python + Flask** and cloud storage for automated report generation.

Projects

Employee Attrition & Performance Analysis | *Python, SQL, Power BI Metabase,*

- Analyzed HRIS data of **100,000+** employee records, identifying a **50.21%** attrition rate influenced by job satisfaction, experience, and roles
- Developed interactive MIS dashboards in **Power BI** & **Tableau**, highlighting critical insights that led to data-driven HR and management decisions

fPI Analyzer: Predictive Analysis of Parkinson's Disease | *Python, Machine Learning, Flask*

- Devised a novel **Frequency Parkinson's Indicator (fPI)** to predict Parkinson's from acoustic sound features with high discriminatory power
- Conducted extensive EDA and trained an **XGBoost** model achieving **96.52% accuracy**, subsequently deploying the solution on **Flask** using **Docker** for accessibility

V-Surveillance: Real-Time Drone-based AI System | *Python, YOLOv12m, DeepSORT*

- Engineered a real-time aerial surveillance pipeline for multi-object detection and tracking on high-resolution drone imagery
- Optimized model inference latency by **44%** (from **210ms** $\rightarrow$ **118ms/frame**) using TensorRT, achieving $\sim$15 FPS real-time performance
- Implemented distributed processing, achieving **92.6% mAP** across 6 object categories, with a visualization dashboard for temporal trajectory analysis

EduSync: Data Analytics for Learning Insights | *Python, PostgreSQL, Tableau*

- Analyzed user interaction events from a **PostgreSQL** database to derive engagement patterns, learning behaviors, and participation trends across cohorts
- Developed interactive dashboards in **Tableau** to visualize key metrics (e.g., active learners, session durations) for stakeholders
- Built predictive models using **Python** (Scikit-learn, XGBoost) to identify at-risk students and forecast participation, enabling proactive learning interventions

Publications

An Enhanced Object-Oriented Programming-Based Web Page Linker — **Apr 2024**
IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI) — *Published*

V-Surveillance: A Hybrid Deep Learning Framework for Real-Time Aerial Surveillance Using Drone Imagery — **Feb 2026**
IEEE International Conference on Information and Communication Technology (CICT) — *Published*

Education

International Institute of Information Technology, Naya Raipur — **78%**
B.Tech in Data Science and Artificial Intelligence — *Nov 2022 - July 2026*

Skills

Programming & Core Tools: Python (Pandas, NumPy, Scipy), SQL (PostgreSQL), R, GraphQL, Git/GitHub, Docker, Linux

ML & Deep Learning: Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, PySpark, OpenCV, Transformers, SciPy

Data Visualization & BI: Power BI, Tableau, Matplotlib, Plotly, Seaborn, Google Analytics, Interactive Dashboards

Cloud & Databases: AWS, Google Cloud (GCP), Microsoft Azure, MongoDB, PostgreSQL, Hadoop, Databricks

Relevant Coursework: Statistical Data Analysis, Deep Learning, Big Data Tools, Data Visualization, DBMS, Data Pre-Processing & Mining

Professional Attributes: Analytical Thinking, Strategic Communication, Problem Solving, Resilience, Leadership, Teamwork